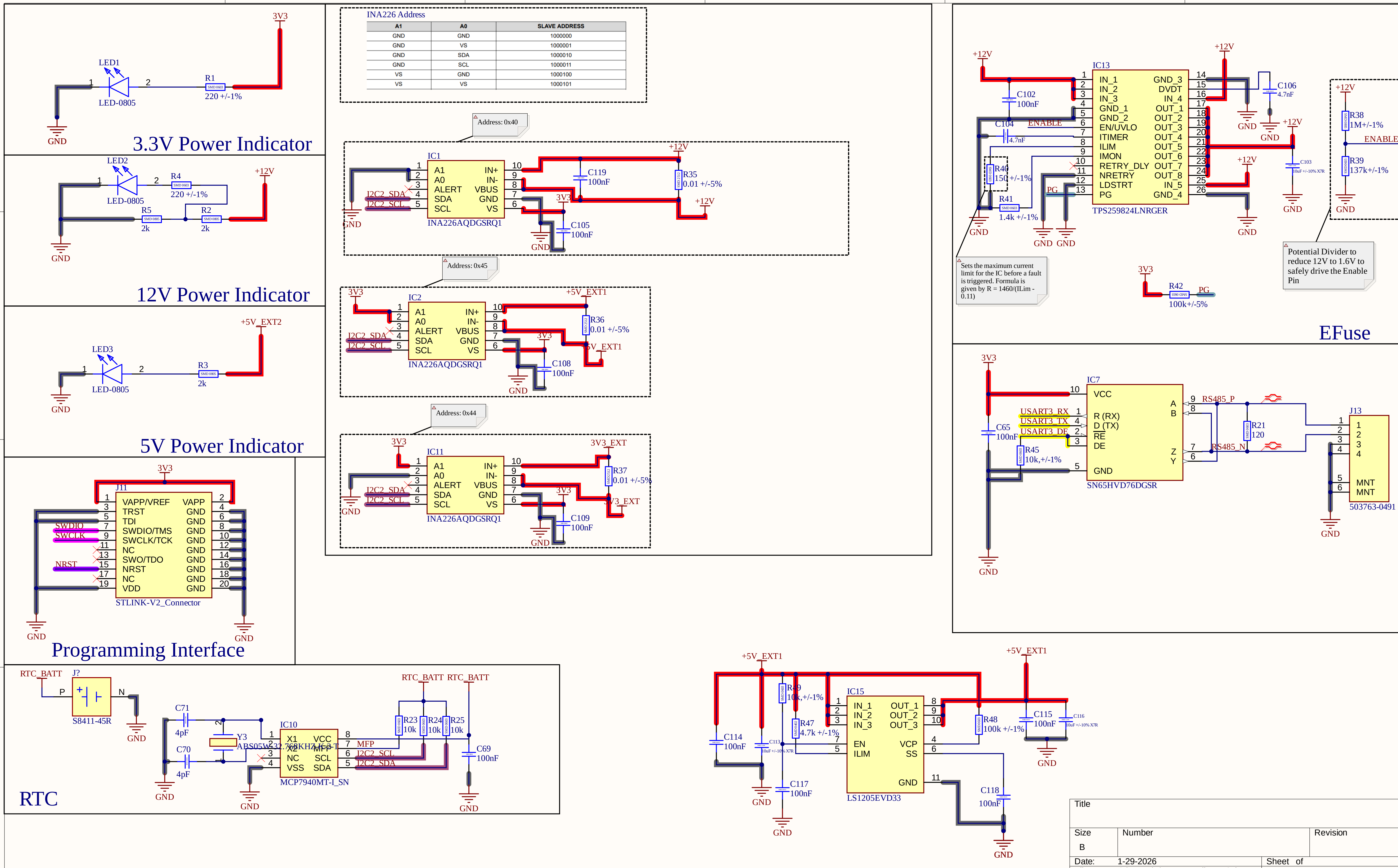

INA226 Address

A1	A0	SLAVE ADDRESS
GND	GND	1000000
GND	VS	1000001
GND	SDA	1000010
GND	SCL	1000011
VS	GND	1000100
VS	VS	1000101

Title

Size: B

Number

Revision

Date: 1-29-2026

Sheet of

File: C:\Users\..\Power_Indicator.SchDoc

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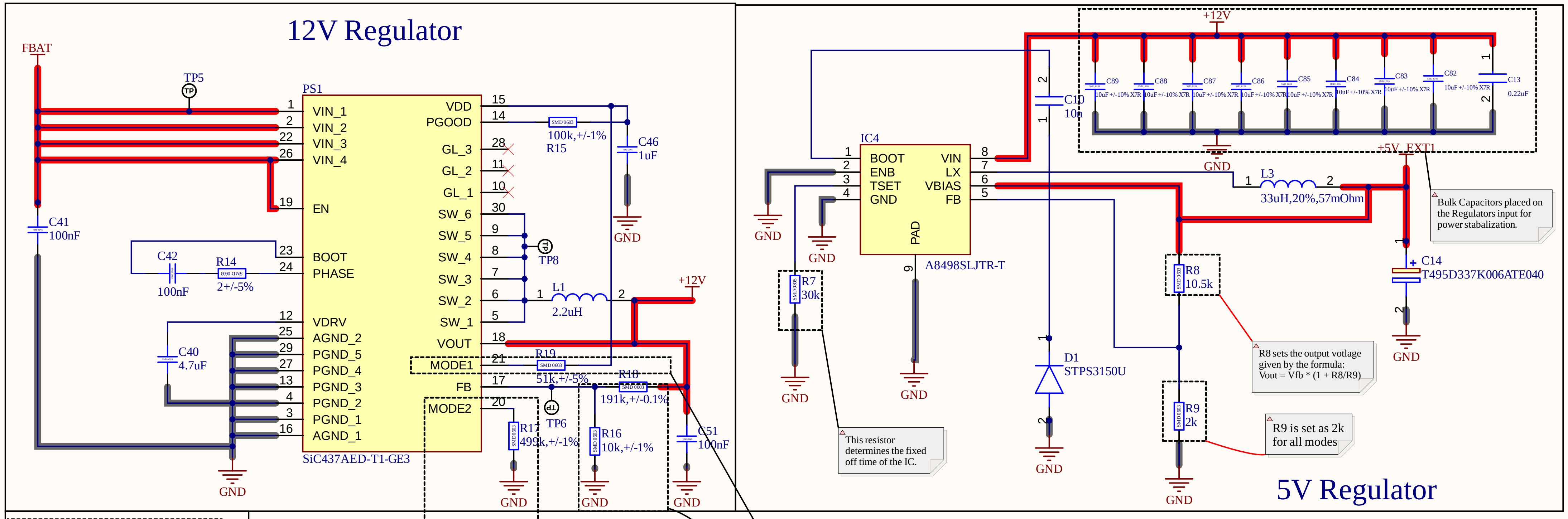
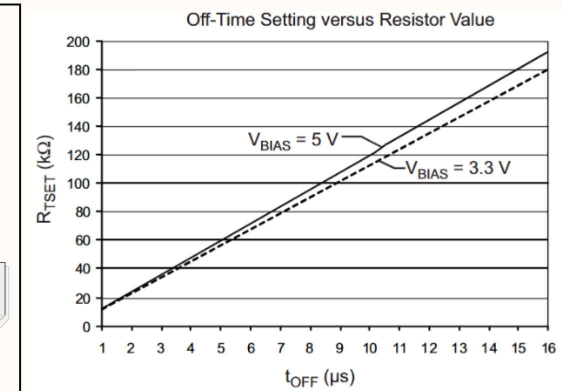


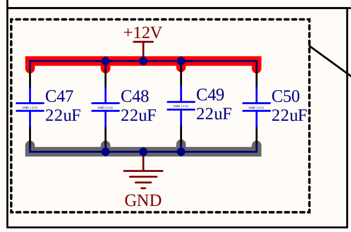
Bulk Capacitors placed on the Regulators input for power stabalization.

R8 sets the output voltage given by the formula: Vout = Vfb * (1 + R8/R9)

R9 is set as 2k for all modes

This resistor determines the fixed off time of the IC.





Output Decoupling according to simulations ran on software provided by manufacturer of the IC.

Mode 2 sets the out current limit to 100% by using 499k ohm resistor. By connecting this pin to Ground. we set the soft-start time to 3ms.

Mode 1 sets the switching frequency to 300kHz and by pulling it high to Vdd the regulator works in constant current mode.

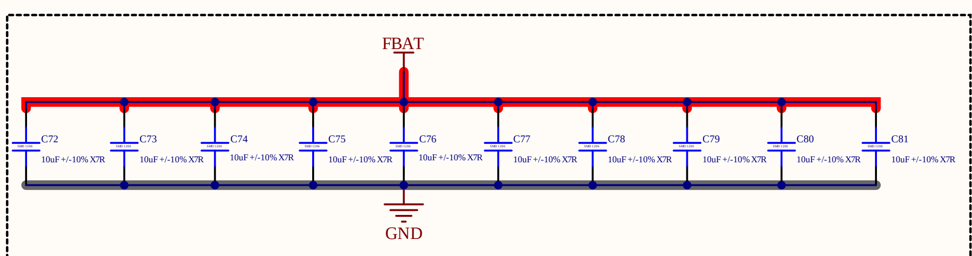
R16 and R18 are used to selected the output voltage from the regulator. The maximum value of R16 is 10k before a output voltage drift occurs. The formula for R18 is given by R16*(Vout - Vfb)/Vfb where Vfb is 0.6V.

TABLE 1 - MODE1 CONFIGURATION SETTINGS

OPERATION	CONNECTION	f_{SWITCH} (kHz)	R_{MODE1} (kΩ)
Skip	to A_{GND}	300	51
		500	100
		750	200
		1000	499
Forced CCM	to V_{DD}	300	51
		500	100
		750	200
		1000	499

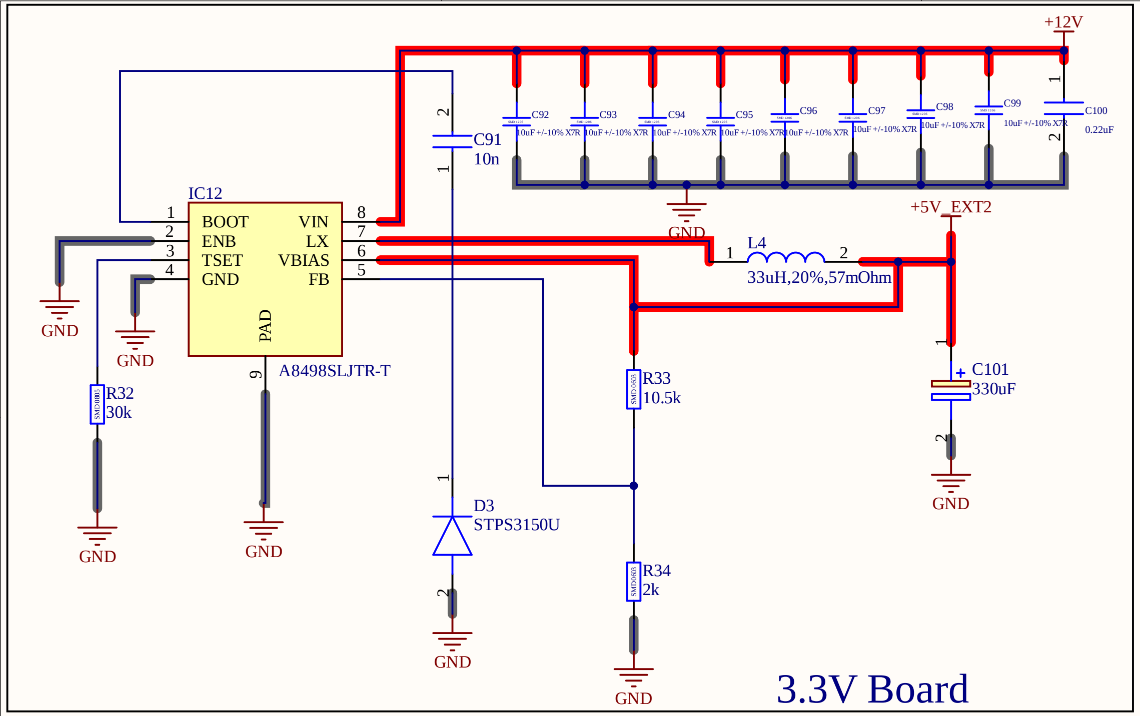
TABLE 2 - MODE2 CONFIGURATION SETTINGS

SOFT-START TIME	CONNECTION	I_{LIMIT} (%)	R_{MODE2} (kΩ)
3 ms	to A_{GND}	30	51
		54	100
		78	200
		100 % (18 A on SiC437) 100 % (12 A on SiC438)	499
6 ms	to V_{DD}	30	51
		54	100
		78	200
		100 % (18 A on SiC437) 100 % (12 A on SiC438)	499



Input Bulk Capacitor used to reduce ripple current

Size B	FCSM No.	DWG No.	Rev
Scale		Sheet	



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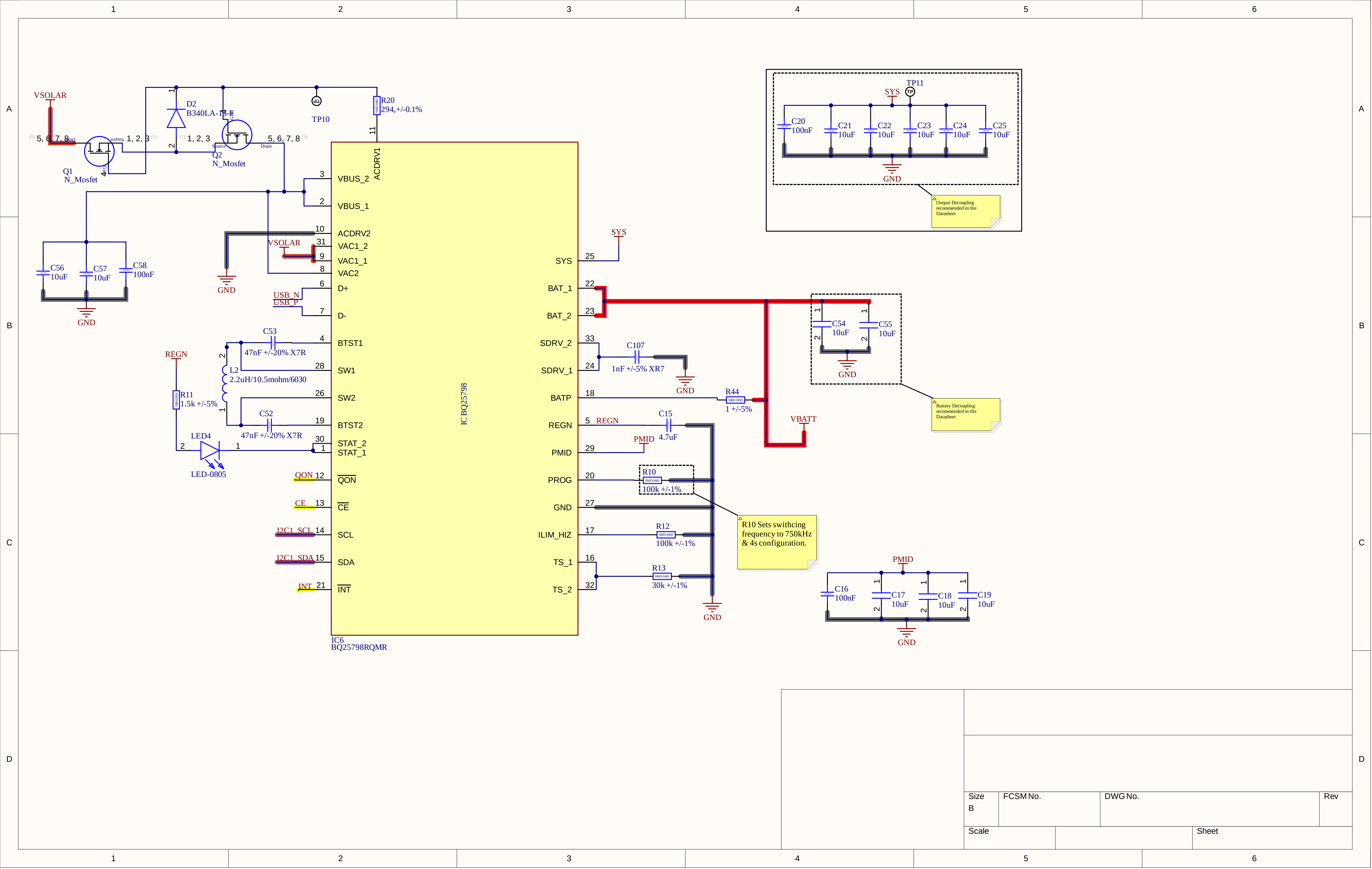
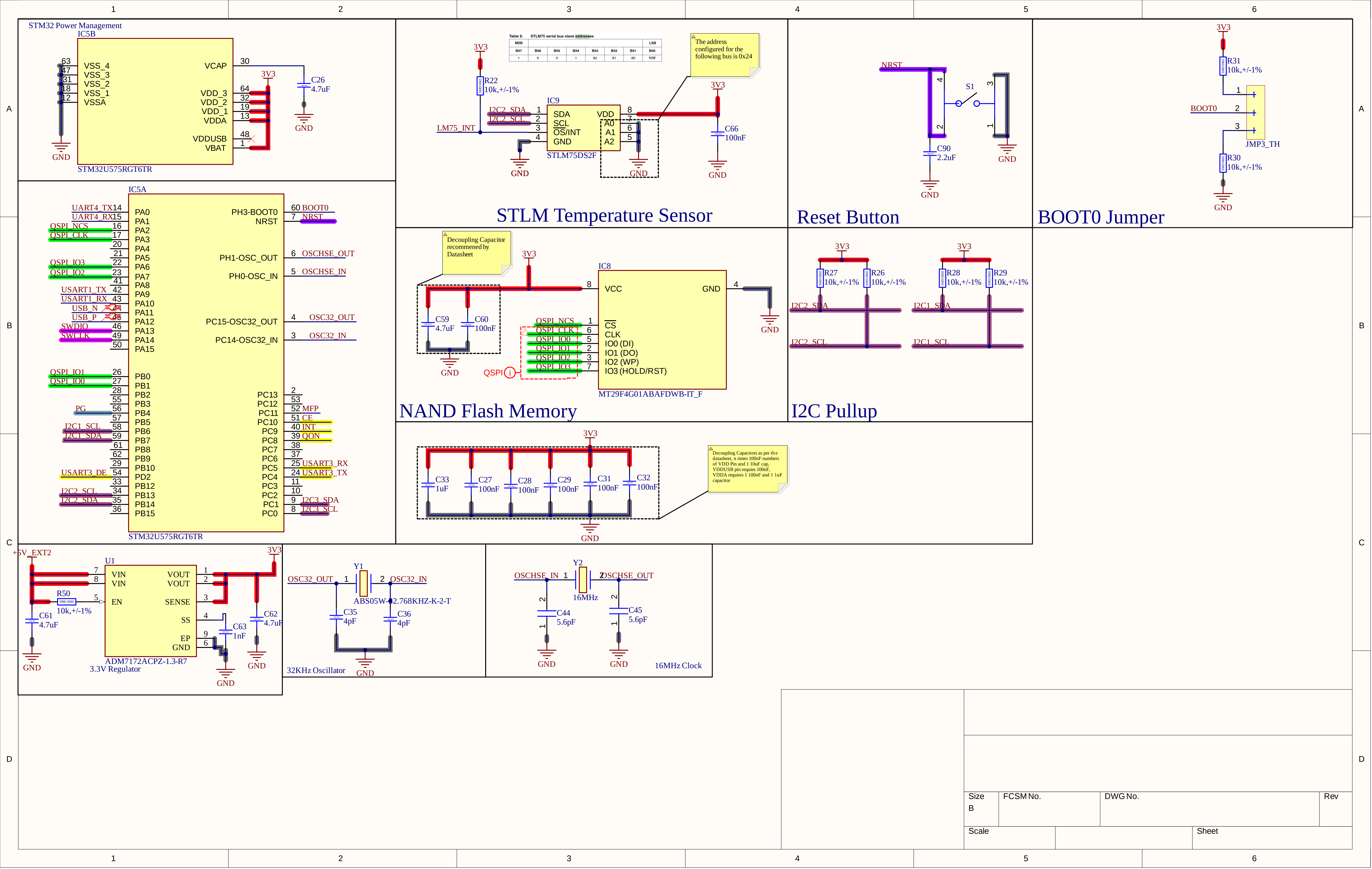
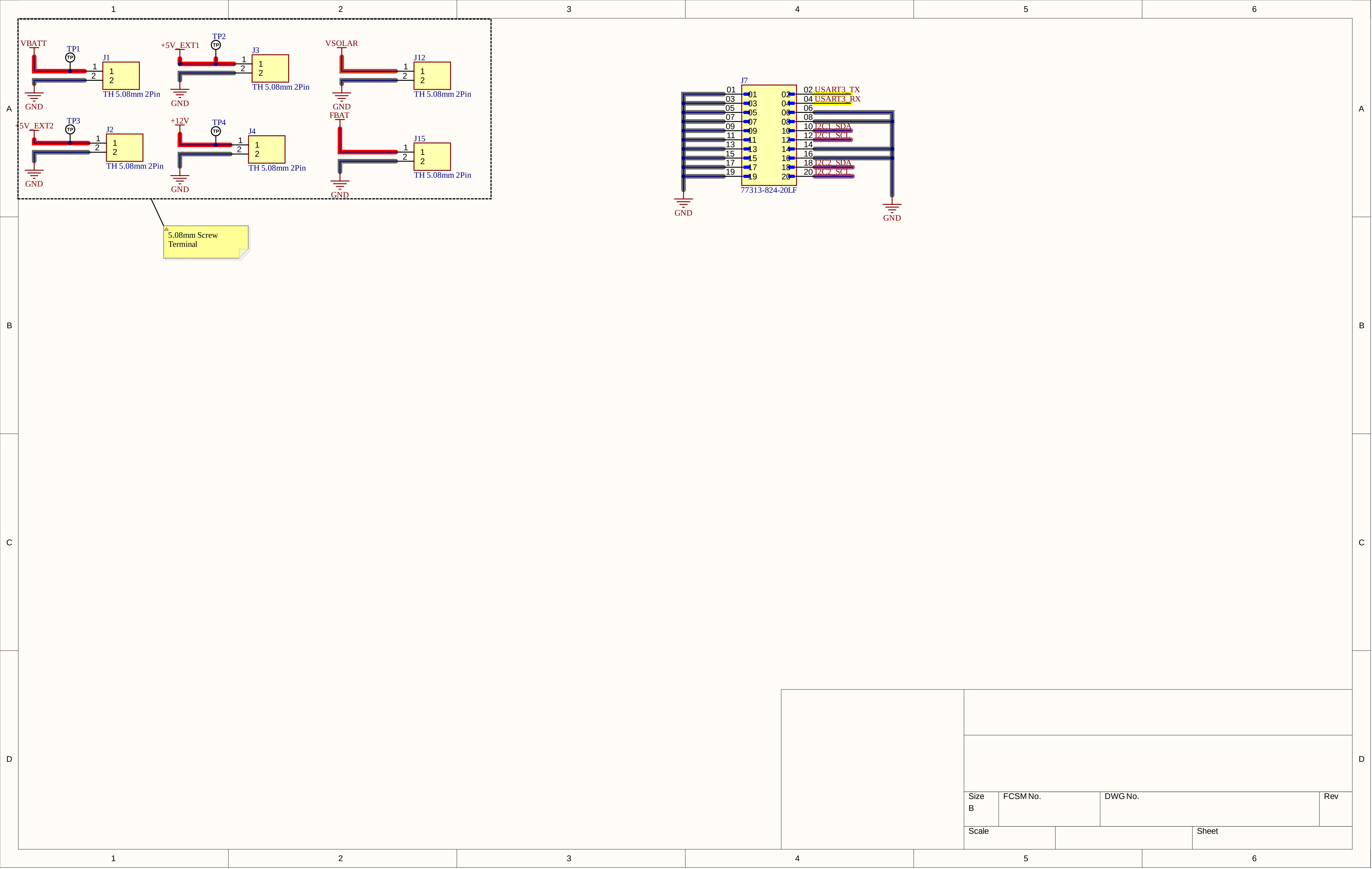


VSOLAR
Q1
N_Mosfet
D2
B340LA-13-F
Q2
N_Mosfet
TP10
R20
294,+/-0.1%
ACDRV1
VBUS_2
VBUS_1
ACDRV2
VAC1_2
VAC1_1
VAC2
D+
D-
BTST1
SW1
SW2
BTST2
STAT_2
STAT_1
QON
CE
SCL
SDA
INT
IC BQ25798
IC6
BQ25798RQMR
SYS
BAT_1
BAT_2
SDRV_2
SDRV_1
BATP
REGN
PMID
PROG
GND
ILIM_HIZ
TS_1
TS_2
C56
10uF
C57
10uF
C58
100nF
GND
USB_N
USB_P
C53
47nF +/-20% X7R
L2
2.2uH/10.5mohm/6030
C52
47nF +/-20% X7R
REGN
R11
1.5k +/-5%
LED4
LED-0805
QON
CE
I2C1_SCL
I2C1_SDA
INT
C107
1nF +/-5% XR7
R44
1 +/-5%
VBATT
C15
4.7uF
R10
100k +/-1%
R12
100k +/-1%
R13
30k +/-1%
R10 Sets swithcing frequency to 750kHz & 4s configuration.
TP11
SYS
C20
100nF
C21
10uF
C22
10uF
C23
10uF
C24
10uF
C25
10uF
Output Decoupling recommended in the Datasheet
C54
10uF
C55
10uF
Battery Decoupling recommended in the Datasheet
PMID
C16
100nF
C17
10uF
C18
10uF
C19
10uF
Size
B
FCSM No.
DWG No.
Rev
Scale
Sheet



STM32 Power Management
IC5B
VSS_4
VSS_3
VSS_2
VSS_1
VSSA
VCAP
VDD_3
VDD_2
VDD_1
VDDA
VDDUSB
VBAT
3V3
C26
4.7uF
GND
STM32U575RGT6TR
IC5A
UART4_TX
UART4_RX
QSPI_NCS
QSPI_CLK
QSPI_IO3
QSPI_IO2
USART1_TX
USART1_RX
USB_N
USB_P
SWDIO
SWCLK
QSPI_IO1
QSPI_IO0
PG
I2C1_SCL
I2C1_SDA
USART3_DE
I2C2_SCL
I2C2_SDA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PD2
PB12
PB13
PB14
PB15
PH3-BOOT0
NRST
PH1-OSC_OUT
PH0-OSC_IN
PC15-OSC32_OUT
PC14-OSC32_IN
PC13
PC12
PC11
PC10
PC9
PC8
PC7
PC6
PC5
PC4
PC3
PC2
PC1
PC0
BOOT0
OSCHSE_OUT
OSCHSE_IN
OSC32_OUT
OSC32_IN
MFP
CE
INT
QON
USART3_RX
USART3_TX
I2C3_SDA
I2C3_SCL
STM32U575RGT6TR
STLM Temperature Sensor
Table 9. STLM75 serial bus slave addresses
The address configured for the following bus is 0x24
R22
10k,+/-1%
IC9
SDA
SCL
OS/INT
GND
VDD
A0
A1
A2
I2C2_SDA
I2C2_SCL
LM75_INT
STLM75DS2F
C66
100nF
Reset Button
NRST
S1
C90
2.2uF
BOOT0 Jumper
R31
10k,+/-1%
R30
10k,+/-1%
JMP3_TH
NAND Flash Memory
Decoupling Capacitor recommened by Datasheet
C59
4.7uF
C60
100nF
IC8
VCC
GND
CS
CLK
IO0 (DI)
IO1 (DO)
IO2 (WP)
IO3 (HOLD/RST)
QSPI
MT29F4G01ABAFDWB-IT_F
I2C Pullup
R27
10k,+/-1%
R26
10k,+/-1%
R28
10k,+/-1%
R29
10k,+/-1%
C33
1uF
C27
100nF
C28
100nF
C29
100nF
C31
100nF
C32
100nF
Decoupling Capacitors as per the datasheet, n times 100nF numbers of VDD Pin and 1 10uF cap, VDDUSB pin requies 100nF, VDDA requires 1 100nF and 1 1uF capacitor
+5V_EXT2
U1
VIN
EN
VOUT
SENSE
SS
EP
R50
10k,+/-1%
C61
4.7uF
C62
4.7uF
C63
1nF
ADM7172ACPZ-1.3-R7
3.3V Regulator
Y1
ABS05W-32.768KHZ-K-2-T
C35
4pF
C36
4pF
32KHz Oscillator
Y2
16MHz
C44
5.6pF
C45
5.6pF
16MHz Clock
Size
B
FCSM No.
DWG No.
Rev
Scale
Sheet



VBATT
TP1
J1
TH 5.08mm 2Pin
GND
+5V_EXT1
TP2
J3
TH 5.08mm 2Pin
GND
VSOLAR
J12
TH 5.08mm 2Pin
GND
+5V_EXT2
TP3
J2
TH 5.08mm 2Pin
GND
+12V
TP4
J4
TH 5.08mm 2Pin
GND
FBAT
J15
TH 5.08mm 2Pin
GND
5.08mm Screw Terminal
J7
77313-824-20LF
USART3_TX
USART3_RX
I2C1_SDA
I2C1_SCL
I2C2_SDA
I2C2_SCL
GND
GND
Size
B
FCSM No.
DWG No.
Rev
Scale
Sheet